20260410_121735

4 minutes ago   32Second Look

$$y\cos(x) + 2xe^{y} + (\operatorname{sen}(x) + x^2e^{y} - 1)\frac{dy}{dx} = 0$$

Exacta.

$$M(x,y) = y\cos(x) + 2xe^{y}$$

$$N(x,y) = \operatorname{sen}(x) + x^2e^{y} - 1$$

$$\frac{\partial M}{\partial y} = \cos(x) + 2xe^{y}$$

$$\frac{\partial N}{\partial x} = \cos(x) + 2xe^{y}$$

Existe una F; $\int \frac{\partial F}{\partial x}dx = \int M dx$ y $\boxed{\frac{\partial F}{\partial y} = N}$

$$F(x,y) = \int (y\cos(x) + 2xe^{y})dx = y\operatorname{sn}(x) + x^2e^{y} + h(y)$$

$$F(x,y) = \int (y\cos(x) + 2xe^{y})dx + h(y)$$

$$= y\operatorname{sen}(x) + x^2e^{y} + h(y)$$

$$\frac{\partial F}{\partial y} = \operatorname{sen}(x) + x^2e^{y} + \frac{dh}{dy}$$

$$\cancel{\operatorname{sen}(x)} + \cancel{x^2e^{y}} - 1 = \cancel{\operatorname{sen}(x)} + \cancel{x^2e^{y}} + \frac{dh}{dy}$$

$$\frac{dh}{dy} = -1; \quad \boxed{h(y) = -y}$$

$$F(x,y) = y\operatorname{sen}(x) + x^2e^{y} - y$$

Solución:

$$\boxed{y\operatorname{sen}(x) + x^2e^{y} - y = K}$$

$$\frac{dy}{dx}\operatorname{sen}(x) + y\cos(x) + 2xe^{y} + x^2e^{y}\frac{dy}{dx} - \frac{dy}{dx} = 0$$

$$y\cos(x) + 2xe^{y} + (\operatorname{sen}(x) + x^2e^{y} - 1)\frac{dy}{dx} = 0$$

$$M = y\cos(x) + 2xe^{y} = \frac{\partial F}{\partial x} \qquad N = \frac{\partial F}{\partial y}$$